Anchor Datasets

CSAI2017P: Applied Machine Learning (Lab)

Reference for lab assignments LA1–LA10

The ten lab assignments reuse three families of dataset. The point is that you learn the methods rather than spending every fortnight wrestling a new file format. Download each one *once* into a local `data/` folder and keep it there for the semester.

Anchor A – Tabular records

Mixed numeric and categorical rows, one record per entity. Used for EDA, regression, classification, scoring and anomaly work.

ID	Dataset	Source	Notes
A1	California Housing	`sklearn.datasets.fetch_california_housing`	20,640 rows, 8 numeric features, numeric target MedHouseVal ($100k units). Ships with scikit-learn; no download.
A2	Telco Customer Churn	`Kaggle: blastchar/telco-customer-churn`	7,043 rows, 21 mixed features, binary target Churn. Download the CSV once and keep it in `data/`.
A3	Statlog German Credit	`OpenML: credit-g (id 31)`	1,000 rows, 20 mixed features, binary target class (good/bad credit). Fetch with `sklearn.datasets.fetch_openml('credit-g', version=1)`.

Anchor B – Images and image-derived features

Pixel grids and features measured from images. Used for image classification, clustering on images and diagnostic tasks.

ID	Dataset	Source	Notes
B1	scikit-learn digits	`sklearn.datasets.load_digits`	1,797 images of 8×8 greyscale handwritten digits, 10 classes. A small stand-in for MNIST that trains in seconds.
B2	MNIST (full)	`sklearn.datasets.fetch_openml('mnist_784')`	70,000 images of 28×28 greyscale digits. Use a subsample of 10,000 unless you have time to spare.
B3	Breast Cancer Wisconsin (Diagnostic)	`sklearn.datasets.load_breast_cancer`	569 rows, 30 features computed from digitised images of fine-needle aspirates, binary target malignant/benign.
B4	PlantVillage (leaf subset)	`Kaggle: emmarex/plantdisease`	Leaf photographs labelled by crop and disease. Use the tomato subset (10 classes) resized to 64×64 to keep runtimes sane.

Anchor C – Sequences and text

Data with an order that matters, or free text. Used for time-series regression, text classification and signal work.

ID	Dataset	Source	Notes
C1	Daily stock OHLCV	`Kaggle: any single-ticker daily history, or yfinance`	One row per trading day with Open/High/Low/Close/Volume. Pick one ticker and at least three years of history.
C2	SMS Spam Collection	`UCI ML Repository: SMS Spam Collection`	5,574 English SMS messages labelled ham/spam. A single tab-separated file.
C3	UCI Human Activity Recognition	`UCI: Human Activity Recognition Using Smartphones`	561 features engineered from accelerometer and gyroscope windows, 6 activity classes, subject-wise train/test split supplied.

Loading snippets

```
# A1 California Housing (no download)
from sklearn.datasets import fetch_california_housing
X, y = fetch_california_housing(return_X_y=True, as_frame=True)

# A3 German Credit (downloads once, then cached)
from sklearn.datasets import fetch_openml
d = fetch_openml("credit-g", version=1, as_frame=True)

# B1 digits (no download)
from sklearn.datasets import load_digits
dig = load_digits() # dig.images 8x8, dig.data 64 features

# B3 Breast Cancer Wisconsin (no download)
from sklearn.datasets import load_breast_cancer
bc = load_breast_cancer(as_frame=True)

# A2 Telco / C2 SMS Spam / C1 stock / B4 leaves : place the CSV in data/
import pandas as pd
df = pd.read_csv("data/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

Reproducibility

Record the exact file name, the download date and the row count of every downloaded dataset in your notebook. Public datasets are revised without notice, and a result you cannot reproduce is a result you cannot defend in the lab quiz.

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.